Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

Student Name	Vicky Pukale
Roll Number	23102A0033
Grade and Subject Teacher's Signature	

Task Number	15
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Final End-to-End Release, Documentation and Viva: Run the complete Loan Application Tracking Portal DevOps workflow from Git commit to Jenkins build, Selenium quality gate, Docker deployment and Puppet/Ansible provisioning. Complete technical documentation, architecture, troubleshooting guide, limitations and future enhancement plan.
Deliverable:	# Final End-to-End Release, Documentation & Viva ## GitHub Link: https://github.com/vickypukale/loan-application-tracking-portal.git ## Final Step: Commit, Tag, and Push the Complete Release: In CMD or PowerShell git add . D:\Vicky_Project\DevOps\DevOps_Project>git add . warning: in the working copy of 'docs/FINAL_TECHNICAL_DOCUMENTATION.md', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'docs/TASK_14_RELIABILITY_AND_ROLLBACK_DELIVERABLE.md', LF will be replaced by CRLF the next time Git touches it warning: in the working copy of 'docs/VIVA_DEFENSE_PREPARATION_GUIDE.md', LF will be replaced by CRLF the next time Git touches it git commit -m "docs: finalize end-to-end technical documentation, architecture guide, and viva defense reference"

```
D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "docs: finalize end-to-end technical documentation, architecture guide, and viva defense reference"
[main 9127d87] docs: finalize end-to-end technical documentation, architecture guide, and viva defense reference
 21 files changed, 335 insertions(+)
 create mode 100644 DOC/DevOps_Task10_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task11_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task12_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task13_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task14_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task6_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task7_Vicky_Pukale_23102A0033.docx
 create mode 100644 DOC/DevOps_Task8_Vicky_Pukale_23102A0033.docx
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 create mode 100644 PDF/DevOps_Task9_Vicky_Pukale_23102A0033.pdf
 create mode 100644 docs/FINAL_TECHNICAL_DOCUMENTATION.md
 create mode 100644 docs/TASK_14_RELIABILITY_AND_ROLLBACK_DELIVERABLE.md
 create mode 100644 docs/VIVA_DEFENSE_PREPARATION_GUIDE.md
```

git tag -a v2.0.0-final -m "Release v2.0.0-final: Complete 15-Task End-to-End DevOps Capstone Implementation"

git push origin main –tags

```
D:\Vicky_Project\DevOps\DevOps_Project>git tag -a v2.0.0-final -m "Release v2.0.0-final: Complete 15-Task End-to-End DevOps Capstone Implementation"

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main --tags
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 16 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (27/27), 12.67 MiB | 4.53 MiB/s, done.
Total 27 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   b8f69a3..9127d87  main -> main
 * [new tag]         v2.0.0-final -> v2.0.0-final
```

Summary of What is Built & Mastered (Tasks 1 to 15):

#	Task Area	What You Built & Achieved
1-3	**Architecture & Planning**	Problem statement, 15-task backlog, DoD, DevOps workflow diagram, and 3-Tier Spring Boot / H2 architecture.
4-6	**Git & Core Feature MVP**	Multi-branching (`main`, `develop`, `feature/*`), PR review, conflict resolution, KPI indicators, search dashboard, status drill-down with timeline, alerts queue, and `v1.0.0` tagging.
7-8	**CI & Pipeline as Code**	Jenkins installation, SCM polling, automated Maven build, and declarative `Jenkinsfile` with parameterized environments (DEV/PROD).
9-10	**Testing & Quality Gates**	Headless Chrome Selenium WebDriver suite testing **4 critical user journeys**, automatic failure screenshots, defect injection, pipeline failure demonstration, and fix verification.
11-12	**Docker & Continuous Delivery**	Multi-stage `Dockerfile`, non-root user (`appuser`), versioned image builds (`loan-portal:build-<N>`), and automated continuous container deployment.
13-14	**Ansible & Site Reliability**	Server specification, Ansible inventory/playbook, **Idempotency verification (changed=0)**, live Actuator health check on **Port 8081**, and sub-15 second disaster recovery rollback.
15	**Final Release & Viva**	Complete technical report, 5-minute live demo script, DORA metrics, and viva exam defense preparation.